

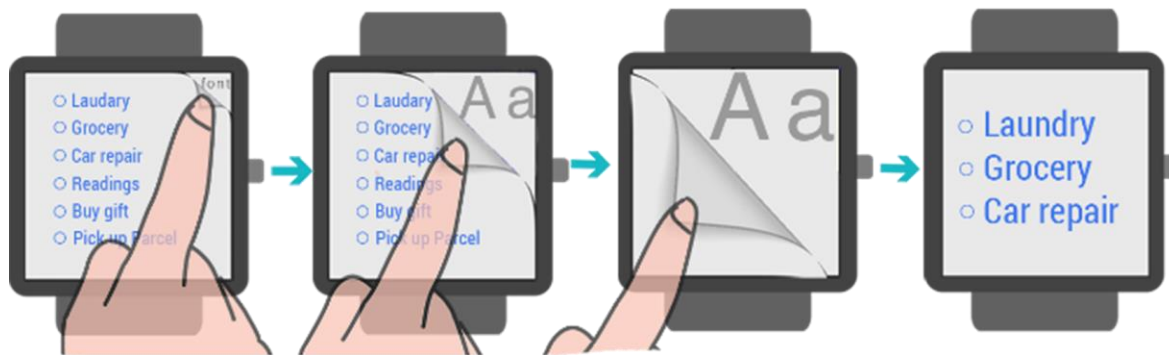
Experiment Design

Today's Lecture

- Experiment design
- Hypothesis
- Identify independent, dependent variables
- Design a controlled experiment

Human-Computer Interaction

- Concerned with the **study, design, implementation** and **evaluation** of human-centric interactive computer systems
- Significant amount devoted to evaluation of novel designs



- E.g., PageFlip, SurroundSee (<https://people.ok.ubc.ca/mkhasan/>)
- But then how do we evaluate systems or design choices?

Types of Evaluation

- Interview
- Questionnaire
- Observation
- Experiment
- Heuristic evaluation
- Cognitive walkthrough

Experiment

- In certain cases you want to make a statement about a particular UI design choice

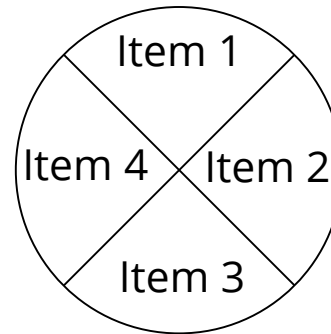
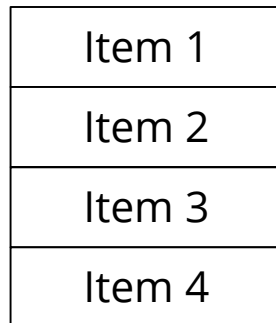
BUTTON

BUTTON

BUTTON

- “I want to know whether the size of buttons contribute to how quickly users click on them”

Or



- “I want to know whether a menu designed in a circular shape (pie menu) is more effective (e.g., time, error) than a regular menu”

Experiment

- Which on-screen keyboard should we use?



Qwerty



Alphabetic



Random

Experiment

- Issues
 - can't just ask people (**preference ≠ performance**)
 - observations alone won't work
 - Cannot control (or even know) all the contributing factors to user's performance
 - i.e. do they use menus more frequently than toolbar buttons because the buttons are too small
- OR
- because the icons are not comprehensible
- OR
- [can go on]

Example

- Is it easier to read with CAPS or without Caps?

THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG

The quick brown fox jumps over the lazy dog

- Want to make a statement whether CAPS are faster to read than non-Caps
- Conclusion would look like:
 - “for text, non-CAPS are 20% faster to read than non-Caps”

Example

- Would it be sufficient to simply show those two slides and do some measurements?
- What are some problems with this kind of setup?
- What would we measure?

THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG

The quick brown fox jumps over the lazy dog

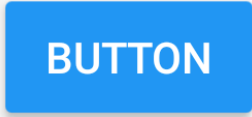
Example

- How do we test this question?

- Need to come up with
 - a hypothesis
 - a set of variables we are going to manipulate
 - a set of variables we are going to measure
 - a task
 - a set of trials
 - ...

Hypothesis

- A statement of the predicted or expected relationship between at least two variables
 - “If I change A, then B will change in this manner...”



BUTTON

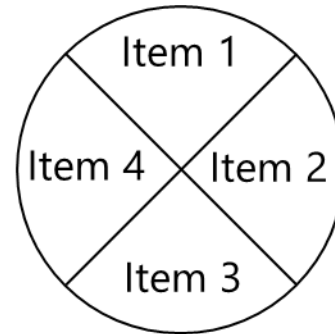
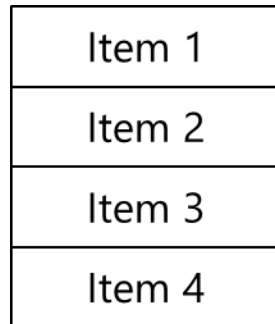


BUTTON

- In experimentation, want a hypothesis to be as specific as possible
 - Makes it easier to test

Hypothesis Testing

- Null hypothesis (H_0)
 - There exists no relationship between manipulating the independent variables and the resultant changes in the dependent variables
- Example:
 - “There is no difference in selection speed between pie-menus and vertical context menus”



- In testing hypothesis, we are seeking to reject the null hypothesis

Hypothesis

- The alternative hypothesis (H_a) claims " H_0 is false"
- H_0 (null hypothesis): There is no difference in the number of cavities in children and teenagers using *crest* and *no-teeth* toothpaste when brushing daily over a one month period
- H_a (alternative hypothesis): children and teenagers using crest toothpaste have fewer cavities than those who use *no-teeth* toothpaste



Independent Variables

- Independent variables are those directly manipulated as part of the experiment
 - the things you manipulate independent of a subject's behavior
- Examples
 - Text type: CAPS or non-caps
 - Menu type: pie or vertical context
- The terms *independent variable* and *factor* are synonymous

Independent Variables

- Hypothesis: there is no difference in the number of cavities in children and teenagers using *crest* and *no-teeth* toothpaste
- What are the independent variables?
 - Toothpaste type: uses Crest or No-teeth toothpaste
 - Age: ≤ 11 years or > 11 years

Independent Variables

- An independent variable (IV) must have at least two levels
- The levels, values, or settings for an IV are the *test conditions*
- Toothpaste Example:
 - Factor: Toothpaste type
 - Levels: Crest or No-teeth toothpaste
 - Factor: Age
 - Levels: ≤ 11 years or > 11 years

Independent Variables

- Hypothesis: There is no difference in user performance (time and error rate) when selecting a single item from a pop-up or a pull down menu of length 3, 6, 9 or 12 items, regardless of the subject's previous expertise in using a mouse

- What are the independent variables? What are the levels?
 - Menu type: pop-up or pull-down
 - Menu length: 3, 6, 9, 12
 - Subject type: expert or novice

Dependent Variables

- Hypothesis includes the dependent variables that will be measured
 - variables dependent on the subject's behaviour / reaction to the independent variable
 - the specific things you set out to quantitatively measure / observe

- Examples
 - Completion time
 - Error rate
 - User preference

Example

- There is no difference in the number of cavities in children and teenagers using glow-right and no-teeth toothpaste when brushing daily over a one month period in toothpaste experiment
- Dependent variable: number of cavities
- Other things we could have measured
 - preference

Example

- There is no difference in user performance (time and error rate) when selecting a single item from a pop-up or a pull down menu of length 3, 6, 9 or 12 items, regardless of the subject's previous expertise in using a mouse
- Dependent variables:
 - time to select an item
 - selection errors made

Experiment

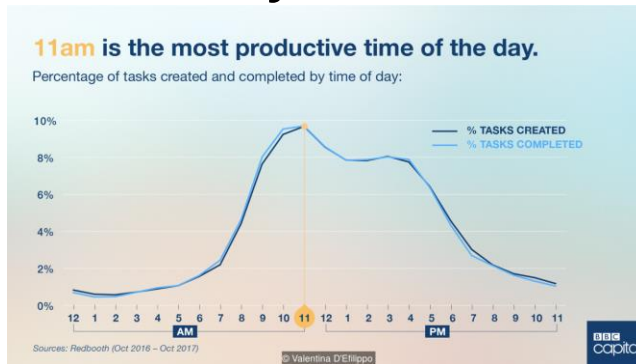
- Common measures
 - Task completion time
 - Error rate
 - User preference
 - Learning rate (novice -> expert transition)
 - Fatigue, comfort?
 - etc.

Control Variable

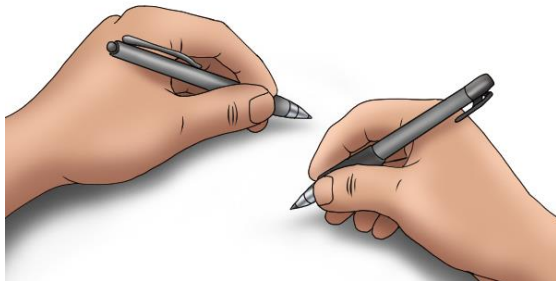
- A control variable is a circumstance (not under investigation) that is kept constant while testing the effect of an independent variable
- Research question: Is there an effect of font color or background color on reading comprehension?
- Independent variables:
 - font color, background color
- Dependent variable:
 - Reading time
- Control variables
 - Font size (e.g., 12 point)
 - Font family (e.g., Times)
 - Ambient lighting (e.g., fluorescent, fixed intensity)
 - Etc.

Nuisance Variables

- Any other factors that can affect the dependent variables
- Examples
 - Time of day



- Handedness



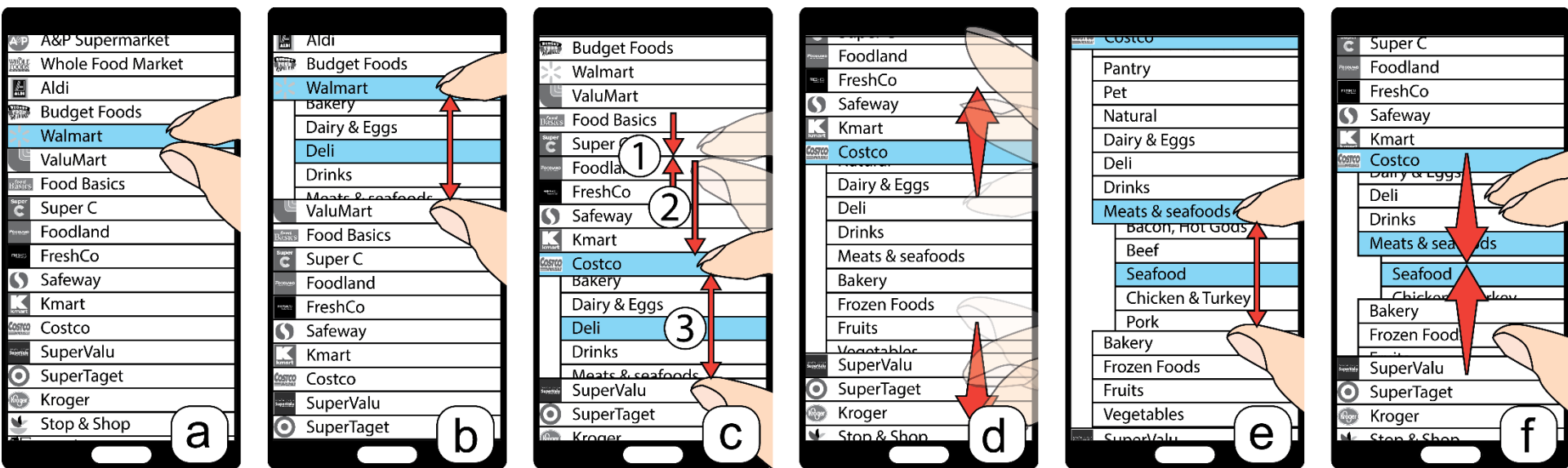
- Goal is to have as few of these as possible

Image Source: <http://www.bbc.com/capital/story/20180102-is-this-the-best-time-for-productivity>
<http://images.mini-ielts.com/images/2017/17/right-and-left-handedness-in-humans.jpg>

Discussion

PinchList: Leveraging Pinch Gestures for Hierarchical
List Navigation on Smartphones

PinchList

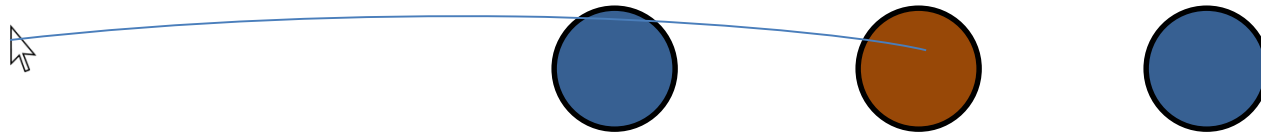


- What are the independent variables? What are the levels?
- What are the dependent variables?

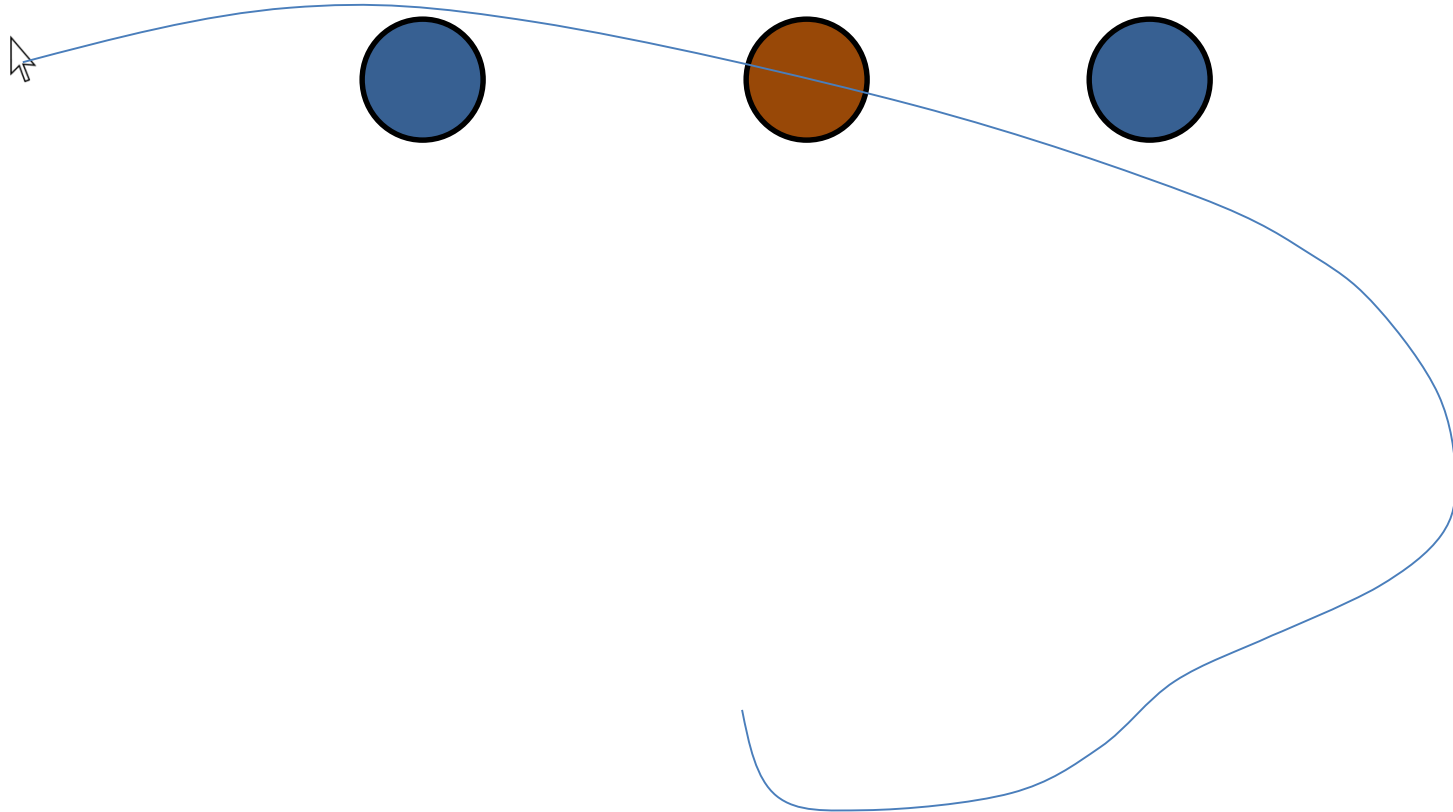
Example

- Comet and Target Ghost: Techniques for Selecting Moving Targets
- Video: <https://www.youtube.com/watch?v=g85ErlsAmUs>
- ACM Digital Library: <https://doi-org.ezproxy.library.ubc.ca/10.1145/1978942.1979065>

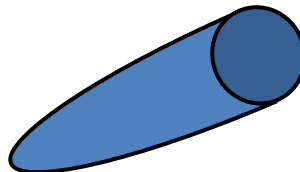
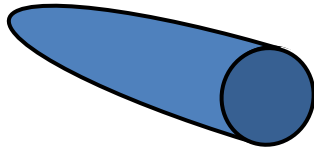
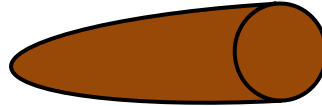
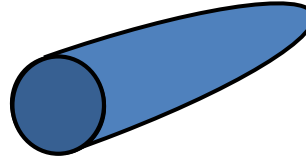
Static Target Selection



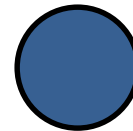
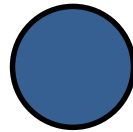
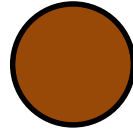
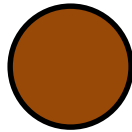
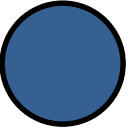
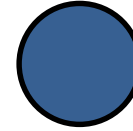
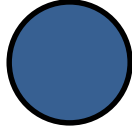
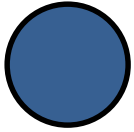
Moving Target Selection



Comet



Target Ghost



Independent Variable: Target Speed



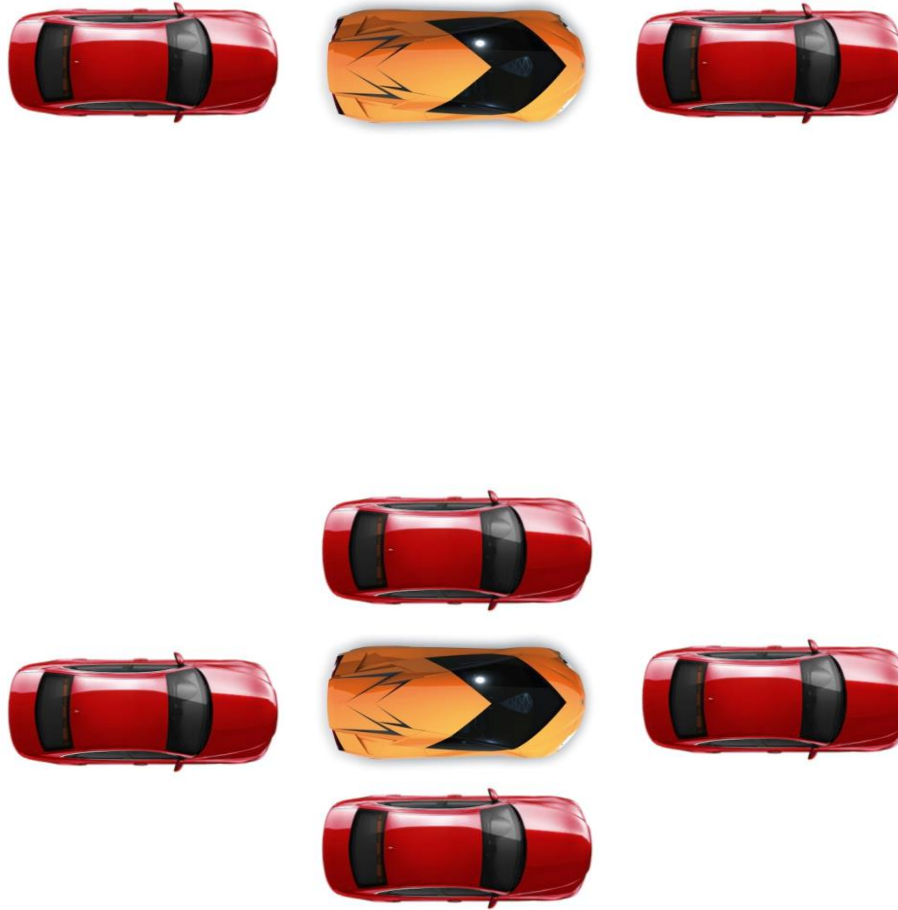
Independent Variable: Path Predictability



Independent Variable: Target Size



Independent Variable: Distracter Density



Participant

- The people participating in an experiment are referred to as participants
 - When referring specifically to the experiment, use the term participants (e.g., “all participants exhibited a high error rate...”)
 - General comments on the problem or conclusions drawn from the results may use other terms (e.g., “these results suggest that users are less likely to...”)

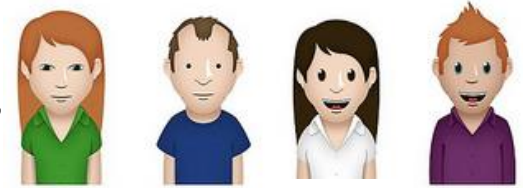
Participant/Subject Selection

- Carefully select and assign subjects to groups
- Ways of controlling subject variability
 - reasonable amount of subjects
 - counterbalancing
 - make different user groups an independent variable
 - screen for anomalies in subject group
 - superstars vs. poor performers



Study Design Decisions

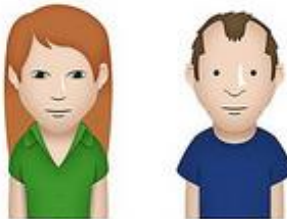
- How will participants be assigned to conditions
- Two options:
 - Between-subjects
 - Within-subjects



Condition 1



Between-subjects



Within-subjects

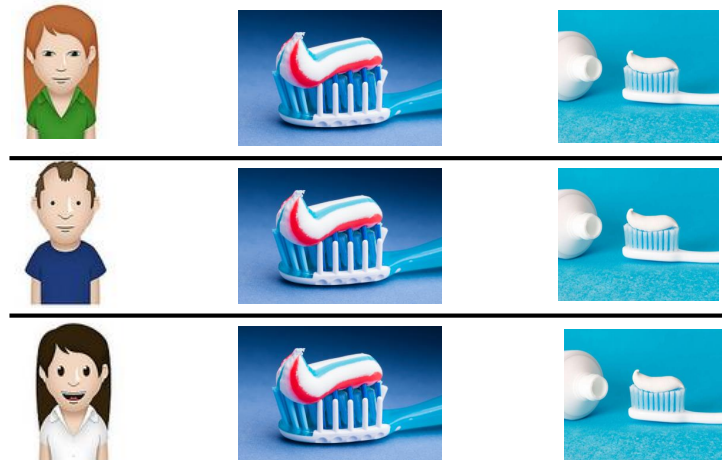


Condition 2



Within-Subjects

- Each participant completes all experimental conditions
- Requires fewer participants
- Allow participants to make direct comparative statements
 - e.g., Technique A is faster than technique B
- Order effects are possible



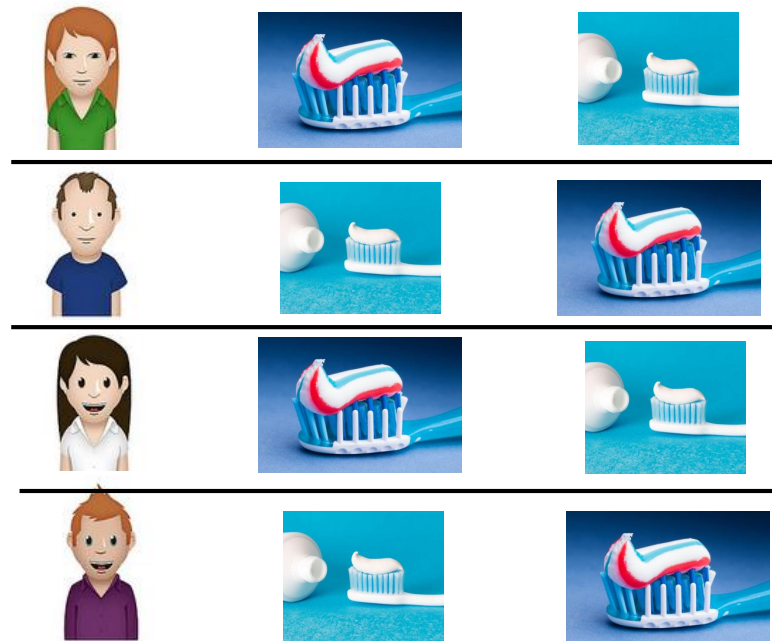
- To account for these, order of conditions are usually counterbalanced

Between-Subjects

- Each participant does one of the experimental conditions
- Need more participants
- No learning effects (good)
- Also known as “randomized experiments”

Order Effects, Counterbalancing

- Only relevant for within-subjects factors
- Order effects offset by counterbalancing:
 - Participants divided into groups
 - Test conditions are administered in a different order to each group
- Order of administering test conditions uses a Latin square



Latin Squares

- A Latin square of order n is an $n \times n$ grid filled with n symbols so that each symbol appears once in each row and column.

2 x 2

A	B
B	A

3 x 3

A	B	C
B	C	A
C	A	B

4 x 4

A	B	C	D
B	C	D	A
C	D	A	B
D	A	B	C

5 x 5

A	B	C	D	E
B	C	D	E	A
C	D	E	A	B
D	E	A	B	C
E	A	B	C	D

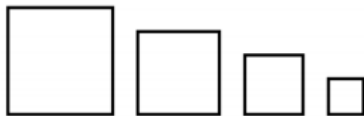
Other Techniques

- Instead of using a Latin square, all orders ($n!$) can be used; 3×3 case $\rightarrow$
- Conditions can be randomized
- Randomizing best if the tasks are brief and repeated often

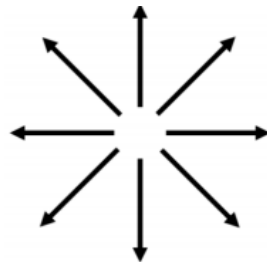
A	B	C
A	C	B
B	C	A
B	A	C
C	A	B
C	B	A

- Examples:

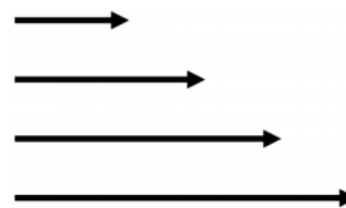
Target size



Movement direction



Movement distance



Experimental Procedure

- Formulate a hypothesis
- Identify independent, dependent variables
- Design a controlled experiment
- Select representative participants
- Assign to conditions
- Run experiment
- Collect data
- Analyze results

Experimental Design

- Participant pool
 - Are the study participants representative of the intended user population?
- The more participants the better
 - Increases confidence of findings
 - Limit?
 - 12 is the most common number of participants in experiments in CHI'12 papers
- Bias
 - Ensure experimenter is not biasing subject

Experiment Procedure

- The experiment procedure encompasses everything that occurs with participants
- The procedure includes the experiment task (obviously), but everything else as well...
 - Arriving, welcoming
 - Signing a consent form
 - Instructions given to participants about the experiment task
 - Demonstration trials, practice trials
 - Rest breaks
 - Administering of a questionnaire or an interview

Data Collection

- Software logging
 - Instrument software to capture detailed timing, error information
- Questionnaires administered before, during and/or after tasks
 - Standardized questionnaires
 - Experimenter-created questionnaires (e.g., Likert scale)
 - E.g., NASA TLX: a measure of workload
- Structured interviews

Please indicate your age: _____

Ratio-scale data

Please indicate your age?

☐ < 20	☐ 20-29	☐ 30-39
☐ 40-49	☐ 50-59	☐ 60+

Ordinal-scale data

Which browser do you use? _____

Open-ended

Which browser do you use?

☐ Mozilla <i>Firefox</i>	☐ Google <i>Chrome</i>
☐ Microsoft <i>IE</i>	☐ Other (_____)

Closed

Questionnaire

Frustration: I felt a high level of insecurity, discouragement, irritation, stress, or annoyance.

1	2	3	4	5	6	7
Strongly disagree			Neutral			Strongly agree

NASA TLX

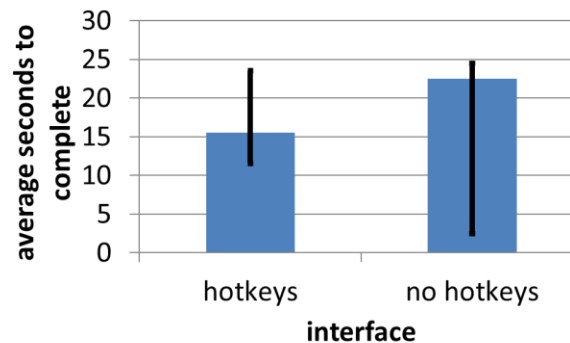
Data Collection

- Planning and pilot testing important
- Ensure conditions are identified, either in the filenames or in the data columns
- Examples:
 - TextInputHuffman-P01-D99-B06-S01.sd2

```
min_keystrokes,keystrokes,presented_characters,transcribed_characters, ...  
55, 59, 23, 23, 29.45, 0, 9.37, 0.0, 2.5652173913043477, 93.22033898305085  
61, 65, 26, 26, 30.28, 0, 10.3, 0.0, 2.5, 93.84615384615384  
85, 85, 33, 33, 48.59, 0, 8.15, 0.0, 2.5757575757575757, 100.0  
67, 71, 28, 28, 33.92, 0, 9.91, 0.0, 2.5357142857142856, 94.36619718309859  
61, 70, 24, 24, 39.44, 0, 7.3, 0.0, 2.9166666666666665, 87.14285714285714
```

Data Analysis

- Different tests include:
 - t-tests, ANOVA, CHI-squared, sign test
- Important to pick the right one for your data



- Statistical Methods for HCI Research
 - <https://yatani.jp/teaching/doku.php?id=hcistats:start>
- Practical Statistics for HCI
 - <http://depts.washington.edu/acelab/proj/ps4hci/index.html>

Data Analysis

- Statistics can use models of randomness to estimate how likely this is by chance alone.
- This estimate is called a p value
 - “the hypothesis that computer experience makes no difference is rejected at the .05 level”
means:
 - a 95% chance that your statement is correct
 - a 5% chance you are wrong

Typical steps to carry out controlled experiments in HCI

- Design
 - State a testable hypothesis
 - Identify independent and dependent variables
 - Design the experimental protocol
 - Choose the user population
- Run
 - Apply for human subjects protocol review
 - Run some pilot participants
 - Modify and finalize the experimental protocol
 - Run the experiment
- Discover
 - Perform statistical analysis
 - Draw conclusion
 - Communicate results

Question?